

nearest whole number?

20. Given that the frequency of $A_3 = 220$ Hz, what is the frequency of A_5 ?

13.5 Math and Sports



Figure 13.15 Fans support their team by attending games and wearing team gear. (credit: "Cheering Touchdowns" by Steven Miller/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Describe why data analytics (statistics) is crucial to advance a team's success.
2. Describe single round-robin method of tournaments.
3. Describe single-elimination method of tournaments.
4. Explore math in baseball, fantasy football, hockey, and soccer (projects at the end of the section).

Sports are big business and entertainment around the world. In the United States alone, the revenue from professional sports is projected to bring in over \$77 billion, which includes admission ticket costs, merchandise, media coverage access rights, and advertising. So, whether or not you enjoy watching professional sports, you probably know someone who does. Some celebrities compete to be part of half-time shows and large companies vie for commercial spots that are costly but reach a staggering number of viewers, some who only watch the half-time shows and advertisements.

Data Analytics (Statistics) Is Crucial to Advance a Team's Success

Analyzing the vast data that today's world has amassed to find patterns and to make predictions for future results has created a degree field for **data analytics** at many colleges, which is in high demand in places that might surprise you. One such place is in sports, where being able to analyze the available data on your team's players, potential recruits, opposing team strategies, and opposing players can be paramount to your team's success.

Hollywood turned the notion of using data analytics into a major motion picture back in 2011 with the release of *Moneyball*, starring Brad Pitt, which grossed over \$110 million. The critically acclaimed movie, based on a true story as shared in a book by Michael Lewis, follows the story of a general manager for the Oakland Athletics who used data analytics to take a team comprised of relatively unheard of players to ultimately win the American League West title in a year's time. The win caught the eye of other team managers and owners, which started an avalanche of other teams digging into the data of players and teams.

In today's world of sports, a team has multiple positions utilizing data analytics from road scouts who evaluate a potential recruit's skills and potential to the ultimate position of general manager who is typically the highest-paid (non-player) employee with the exception of the coaches. Being able to understand and evaluate the available data is big business and is a highly sought after skill set. In college and professional sports, it is no longer sufficient to have a strong playbook and great players. The science to winning is in understanding the math of the data and using it to propel your team to excelling.

Single Round-Robin Tournaments

8 Team round robin

Team 1	Team 2	Team 3	Team 4			
W	L	W	L			
Team 5	Team 6	Team 7	Team 8			
W	L	W	L			
Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Round 7
2 vs 1	3 vs 4	6 vs 2	7 vs 5	1 vs 3	4 vs 6	7 vs 3
3 vs 8	1 vs 7	7 vs 8	8 vs 4	4 vs 2	8 vs 1	8 vs 2
4 vs 7	8 vs 6	4 vs 1	2 vs 3	5 vs 8	2 vs 7	1 vs 5
5 vs 6	2 vs 5	5 vs 3	6 vs 1	6 vs 7	3 vs 6	6 vs 4

Figure 13.16 Single-Round Robin Tournament

A common tournament style is single round-robin tournaments (Figure 13.16), where each team or opponent plays every other team or opponent, and the champion is determined by the team that wins the most games. Ties are possible and are resolved based on league rules.

An advantage of the round-robin tournament style is that no one team has the advantage of **seeding**, which eliminates some teams from playing against each other based on rank of their prior performance. Rather, each team plays every other team, providing equal opportunity to triumph over each team. In this sense, round-robin tournaments are deemed the fairest tournament style.

One hindrance to employing a round-robin-style tournament is the potential for the number of games involved in tournament play to determine a winner. Determining the number of games can be found easily using a formula which, as we will see, can quickly grow in the number of games required for a single round-robin tournament.

FORMULA

The number of games in a single round-robin tournament with n teams is $n(n - 1)/2$.

EXAMPLE 13.22

Calculating the Number of Games in Single Round-Robin Tournaments

Find the number of games in a single round-robin tournament for each of the following numbers of teams:

- 4 teams
- 8 teams
- 20 teams

Solution

- Using the [formula](#) with 4 teams yields $4(4 - 1)/2 = 6$ tournament games.
- Using the [formula](#) with 8 teams yields $8(8 - 1)/2 = 28$ tournament games.
- Using the [formula](#) with 20 teams yields $20(20 - 1)/2 = 190$ tournament games.

YOUR TURN 13.22

Find the number of games in a single round-robin tournament for each of the following numbers of teams:

1. 5 teams
2. 12 teams
3. 25 teams

As the examples show, single round-robin tournament play can quickly grow in the number of games required to determine a champion. As such, some tournaments elect to employ variations of single round-robin tournament play as well as other tournament styles such as elimination tournaments.

8 Team single elimination

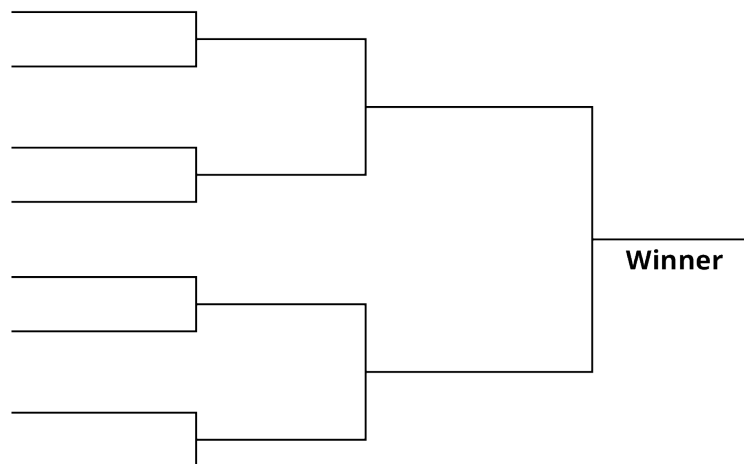


Figure 13.17 Single-Elimination Tournament (credit: final TK)

Single-Elimination Tournaments

When desiring a more efficient tournament style to determine a champion, one option is single-elimination tournaments (Figure 13.17), where teams are paired up and the winner advances to the next round of play. The losing team is defeated from tournament play and does not advance in the tournament, although some leagues offer consolation matches.

FORMULA

The number of games in a single-elimination tournament with n teams is $(n - 1)$.

EXAMPLE 13.23

Calculating the Number of Games in Single-Elimination Tournaments

Find the number of games in a single-elimination tournament for each of the following numbers of teams:

1. 4 teams
2. 8 teams
3. 20 teams

Solution

1. Using the [formula](#) with 4 teams yields $(4 - 1) = 3$ tournament games.
2. Using the [formula](#) with 8 teams yields $(8 - 1) = 7$ tournament games.
3. Using the [formula](#) with 20 teams yields $(20 - 1) = 19$ tournament games.

YOUR TURN 13.23

Find the number of games in a single-elimination tournament for each of the following numbers of teams:

1. 5 teams
2. 12 teams
3. 25 teams

A single-elimination tournament offers an advantage over single round-robin tournament style of play in the number of games needed to complete the tournament. As you can see, in comparing the number of games in a single round-robin tournament in [Example 13.22](#) with the number of games in single-elimination tournament as shown in [Example 13.23](#), the number of games required for single round-robin can quickly become unmanageable to schedule.

There are modifications to both the round-robin and elimination tournament styles such as double round-robin and double-elimination tournaments. Next time you observe a college or professional sporting event, see if you can determine the tournament style of play.

?? WHO KNEW?

Sports Popularity Shifts

Sport has been a popular entertainment venue for hundreds of years and the popularity of various sports shifts over time as well as in different regions of the world today. In today's world, the most popular sport is, overwhelmingly, soccer, with over 4 billion fans followed by cricket with 2.5 billion fans. It may surprise you to learn that American football doesn't rank in the top 10 most popular sports in the world today, yet table tennis ranks in sixth place and golf fills the last spot in the top 10 world sports.

In the early 1930s, baseball ranked a close second, with basketball virtually tying for third place. By the mid-1940s, hockey slightly led in first place over basketball. Ten years later, hockey remained in the number one sport, but cricket pushed basketball to third place. In the 1960s, soccer dominated in popularity. Over the next 30 years, basketball dropped in popularity and there was much movement in the popularity of sports. By the late 1990s, soccer swept to first place, where it has since continued to grow in popularity.

Check Your Understanding

18. How is data analytics used in sports today?
19. What is the most popular sport in today's world?
20. What is the formula to compute the number of games played in a single round-robin tournament with n teams assuming single round?
21. What is the formula to compute the number of games played in a single-elimination tournament?



SECTION 13.5 EXERCISES

1. Briefly describe data analytics.
2. Name a sport's career title that relies on analysis of data.

For the following exercises, compute how many games would be played in the style of tournament and number of teams given in each question. Assume all tournaments are single round-robin or single-elimination.

3. How many games would be played with 4 teams using a round-robin tournament?
4. How many games would be played with 8 teams using a round-robin tournament?
5. How many games would be played with 10 teams using a round-robin tournament?
6. How many games would be played with 25 teams using a round-robin tournament?
7. How many games would be played with 12 teams using a round-robin tournament?
8. How many games would be played with 4 teams using a single-elimination tournament?
9. How many games would be played with 9 teams using a single-elimination tournament?
10. How many games would be played with 15 teams using a single-elimination tournament?
11. How many games would be played with 50 teams using a single-elimination tournament?
12. How many games would be played with 24 teams using a single-elimination tournament?